

Notes: Puckett and Yan (JF, 2011)

The Interim Trading Skills of Institutional Investors

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1 Big picture

- **Question.** Do institutional investors have trading skill within the quarter, and do standard quarterly holdings data miss that skill?
- **Core data innovation.** The paper uses proprietary ANcerno transaction-level data from 1999–2005. Unlike 13F or quarterly mutual-fund holdings, these data identify actual trade dates, trade directions, execution prices, trading volume, fund identifiers, and institution identifiers.
- **Main object.** The paper studies *interim trading performance*: the abnormal performance of stocks bought versus stocks sold from each trade’s execution price through the end of the quarter.
- **Why interim trades matter.** Quarterly holdings miss intraquarter round trips and mis-time trades by assuming they happen at quarter-end. This matters because informed trading opportunities may be short-lived.
- **Headline finding.** Institutional buys outperform institutional sells within the quarter. After commissions, buy-minus-sell interim performance is positive and statistically significant.
- **Persistence.** Funds with high interim trading performance continue to outperform in later quarters. The top quintile remains positive for the next four quarters, while the bottom quintile remains weak.
- **Quarterly-data contrast.** When the authors reconstruct “implied quarterly trades” from net quarterly trading positions, the estimated trading performance becomes negative or insignificant and persistence largely disappears.
- **Source of skill.** Skill is strongest in stocks where information asymmetry or limits to arbitrage are likely to be high: small stocks, high idiosyncratic volatility stocks, illiquid stocks, and high quoted-spread stocks.
- **Liquidity-provision test.** High-skill funds do not earn abnormal returns by providing liquidity. They actually incur higher implicit trading costs, suggesting that they demand liquidity to trade on information.
- **Economic magnitude.** Average after-commission interim trading skill contributes about 20–26 basis points per year to the average portfolio under a turnover assumption; for high-skill funds, the contribution is much larger.
- **Takeaway.** Quarterly holdings data understate short-horizon institutional skill because they are too coarse to capture timing, prices, and round-trip trades.

2 Incremental contribution

- **Moves beyond quarterly holdings.** Earlier studies usually infer institutional trades from changes in quarterly holdings. This paper shows that this proxy can reverse conclusions because it ignores within-quarter timing and round-trip trades.
- **Direct trade-level evidence.** The paper contributes transaction-level evidence on institutional trading skill using exact execution prices and dates. This is stronger than evidence based on inferred trades from 13F holdings or TAQ algorithms.
- **New performance dimension.** Prior research emphasizes long-horizon stock-picking, holdings-based alpha, or market timing. This paper isolates *interim trading skill*: the ability to exploit short-lived opportunities during the quarter.
- **Persistence of short-horizon skill.** The paper documents that interim trading performance is persistent and that persistence is driven by high-skill funds, contrasting with some evidence that mutual-fund persistence is weak or driven by poor performers.
- **Measurement-warning contribution.** A major contribution is methodological: if researchers use coarse quarterly holdings data, they may incorrectly infer that institutions have no skill or negative skill even when trade-level data show positive skill.

- **Skill-source contribution.** The paper links skill to stock environments with high information asymmetry and limits to arbitrage, such as high-IVOL and illiquid stocks.
- **Liquidity-cost contribution.** By showing that high-skill funds pay higher implicit trading costs, the paper rules out a simple liquidity-provision explanation and points to costly informed trading.
- **Policy contribution.** The results matter for disclosure policy: more frequent institutional portfolio disclosure could erode short-lived information advantages that institutions currently exploit.

3 Data and measurement

3.1 ANcerno institutional trading data

- The sample covers institutional equity trades from January 1, 1999 to December 31, 2005.
- ANcerno clients include pension plan sponsors and money managers. The database is designed for execution-cost consulting rather than investment-performance evaluation.
- The final fund-level sample contains 3,816 funds: 3,589 pension funds and 227 money manager funds.
- The sample includes only common stocks with CRSP share codes 10 or 11.
- The data identify trade direction, share volume, dollar volume, execution date, execution price, fund identifier, and institution identifier.

Interpretation: The data are unusually powerful because the paper observes the precise transaction that generates the position, not just the position at a reporting date.

3.2 Stock characteristics and benchmarks

- The paper uses CRSP returns, share price, shares outstanding, trading volume, quoted spreads from TAQ, and book equity from Compustat.
- Abnormal performance is measured using Daniel, Grinblatt, Titman, and Wermers (DGTW) characteristic-adjusted benchmark returns.
- Stock-characteristic sorts use market capitalization, book-to-market, lagged return, turnover, idiosyncratic volatility, Amihud illiquidity, and quoted spread.
- Characteristics are measured at the end of the previous quarter to avoid contemporaneous look-ahead bias.

Interpretation: DGTW adjustment means the performance is not simply compensation for size, value, or momentum exposures. The characteristic splits ask where skill is strongest.

3.3 Round-trip trades

- A round-trip trade occurs when a fund buys and then sells, or sells and then buys, the same stock within the quarter.
- Round-trip trades account for nearly a quarter of all trades in the sample.
- Performance is the holding-period return between the first and second leg of the round trip, adjusted by the DGTW benchmark.

Interpretation: Round-trip trades are the most direct evidence of short-horizon trading opportunities, but they may overstate skill if funds are more likely to close winning trades than losing trades.

3.4 Interim trading performance

For each trade j by fund i in quarter q , define an abnormal holding-period return from the trade execution price to the end of the quarter:

$$AR_{ijq} = R_{ijq}^{exec \rightarrow qend} - R_{DGTW,ijq}^{exec \rightarrow qend}.$$

For each fund-quarter, compute average abnormal returns separately for buys and sells:

$$ITP_{iq} = \overline{AR}_{iq}^{Buy} - \overline{AR}_{iq}^{Sell}.$$

The paper reports both equal-weighted and principal-weighted versions.

Interpretation: A positive value means the stocks a fund buys outperform the stocks it sells during the same quarter, after benchmark adjustment.

3.5 Implied quarterly trading performance

- To mimic quarterly holdings studies, the paper aggregates all trades in a stock within a quarter into a net position.
- Positive net trades are treated as buys; negative net trades are treated as sells.
- Performance is measured over the subsequent quarter as in holdings-based studies.

Interpretation: This deliberately discards trade timing and round-trip information. The contrast with interim performance shows what quarterly data miss.

3.6 Implicit trading costs

For a buy trade, implicit trading cost is execution price minus same-day VWAP. For a sell trade, it is same-day VWAP minus execution price. Costs are scaled by the execution price and expressed as percentages.

Interpretation: If skilled funds merely provided liquidity, they should have lower or negative costs. The paper finds the opposite, which favors an informed-demand interpretation.

4 Models and empirical specifications

4.1 Round-trip performance

For a round-trip trade with first-leg price P_0 , second-leg price P_1 , and DGTW benchmark return R_b , the abnormal return is approximately

$$AR^{RT} = \frac{P_1 - P_0}{P_0} - R_b$$

for buy-then-sell trades, with the sign adjusted for sell-then-buy trades.

Interpretation: Round-trip returns test whether institutions can enter and exit the same stock profitably inside a quarter.

4.2 Interim trading performance for all trades

The main performance measure is

$$ITP_{iq} = \sum_{j \in Buy_{iq}} w_{ijq} AR_{ijq} - \sum_{j \in Sell_{iq}} w_{ijq} AR_{ijq},$$

where w_{ijq} is either an equal weight or a principal weight.

Interpretation: Principal weights better approximate the dollar return earned by the fund. Equal weights better reveal whether skill is broad across trades.

4.3 Persistence regressions and portfolio sorts

Each quarter, funds are sorted into quintiles by current principal-weighted interim performance. The paper then computes subsequent-quarter performance for the same quintiles:

$$ITP_{q+h}^{Q5} - ITP_{q+h}^{Q1}, \quad h = 1, 2, 3, 4.$$

Interpretation: Persistence is the central test of skill. Random luck should not produce a stable ranking across future quarters.

4.4 Quarterly proxy comparison

The implied quarterly performance measure is

$$IQT P_{iq} = \overline{AR}_{i,q+1}^{NetBuy} - \overline{AR}_{i,q+1}^{NetSell},$$

where net buys and net sells are inferred from net quarter-end positions.

Interpretation: This measure asks whether the same sample looks skilled if one only observes the kind of quarterly net-trade proxy available in standard holdings data.

4.5 Stock-characteristic mechanism tests

The paper estimates performance separately for trades in stocks sorted by characteristics:

$$ITP_{iq}^{HighZ} - ITP_{iq}^{LowZ},$$

where Z is size, book-to-market, lagged return, turnover, IVOL, illiquidity, or quoted spread.

Interpretation: Skill should be stronger where private information is more valuable and arbitrage is more difficult, but these are also the stocks with higher trading costs.

4.6 Implicit cost test

Funds are sorted by interim performance quintiles and implicit cost is compared across groups:

$$Cost^{Q5} - Cost^{Q1}.$$

Interpretation: A positive cost spread means high-performance funds pay more to trade. This supports costly information acquisition rather than liquidity provision.

5 Identification and empirical design

5.1 Core identification problem

The empirical challenge is to distinguish actual trading skill from benchmark exposure, timing artifacts, liquidity provision, and sample selection. The authors address this by observing actual trade execution dates and prices, adjusting returns with DGTW benchmarks, comparing buys to sells within fund-quarters, and benchmarking against an intentionally coarse quarterly-trade proxy.

5.2 Why trade-level data identify interim skill

- Trade-level data observe when the fund actually acts.
- Execution prices replace the assumption that all trades occur at quarter-end.
- Round trips are observed rather than netted out.
- Buy and sell performance can be evaluated symmetrically within the same quarter.

Interpretation: If private information is short-lived, trade timing is not a small detail; it is the object being measured.

5.3 Why buy-minus-sell is useful

The buy-minus-sell comparison controls for fund-quarter conditions that affect both sides of the book. If a fund buys stocks that outperform and sells stocks that underperform within the same period, the difference provides a cleaner performance measure than examining buys alone.

5.4 Why persistence matters

One-quarter abnormal performance can reflect luck. Persistence across future quarters is harder to reconcile with pure noise. The paper's persistence results therefore provide stronger evidence of skill than the unconditional average performance result.

5.5 Why quarterly reconstruction is a strong diagnostic

The implied quarterly analysis applies a standard low-frequency method to the same sample. The sharp difference between interim performance and implied quarterly performance is not driven by sample differences; it is driven by measurement frequency.

5.6 Threats to interpretation

- ANcerno clients may differ from the population of institutions.
- The data do not contain fund fees or actual investor-level fund returns.
- Some trades may be flow-driven or rebalancing-driven rather than information-driven.
- DGTW adjustment may not fully remove all risk exposure.
- High implicit costs may partly reflect urgency or market impact, not only private information.

Interpretation: These limitations mean the paper documents trading performance, not necessarily investor net alpha after fees. Still, the trade-level evidence is much cleaner than quarterly holdings proxies for the question it asks.

6 Section-by-section study guide

- **Introduction.** Explains why prior evidence on institutional skill is mixed and why quarterly holdings are too coarse.
- **Data.** Describes ANcerno and explains why the sample is suited to trade-level skill tests.
- **Methodology.** Defines round-trip trades, interim trading performance, DGTW adjustment, and persistence portfolios.
- **Results: round trips.** Shows strong abnormal performance for within-quarter reversals.
- **Results: all trades.** Shows positive interim buy-minus-sell performance even after commissions.
- **Persistence.** Shows that high-skill funds continue to outperform.
- **Quarterly proxy.** Demonstrates that net quarterly trades yield the wrong inference.
- **Mechanisms.** Shows performance is strongest in stocks with high information asymmetry and trading frictions.
- **Conclusion.** Connects the results to disclosure policy and the broader investment-skill debate.

7 Tables and figures

7.1 Table I: Descriptive Statistics for ANcerno Institutional Trading Data and Stock Characteristics + Table II: Performance of Intraquarter Round-Trip Trades

Read:

- Table I shows that the ANcerno sample is large and economically meaningful: thousands of funds, hundreds of institutions, trillions of dollars of annual trading volume, and millions of trades.
- Money manager funds are far fewer than pension funds but account for a large fraction of dollar trading volume.
- The traded stocks are not tiny microcaps; average market capitalization is in the billions, but spreads, illiquidity, and IVOL still vary substantially.
- Table II shows that round-trip trades earn large abnormal returns. DGTW-adjusted returns after commissions are positive for all funds and remain positive for both pension and money-manager subgroups.
- Round-trip persistence is very strong: the difference between top and bottom quintiles remains large in subsequent quarters.

Detailed interpretation: Table I establishes sample scale and helps address external-validity concerns. Table II provides the first direct evidence that institutions can trade profitably within quarters, but round-trip trades may be selected because winners are more likely to be closed quickly. This is why Table III extends the analysis to all trades.

Economic message: The data are rich enough to study institutional trading at the level where information is actually used. Round trips suggest short-lived trading opportunities, but all-trade measures are needed to rule out winner-selection bias.

Table I
Descriptive Statistics for ANcerno Institutional Trading Data and Stock Characteristics

Institutional trading data are obtained from ANcerno Ltd., and the trades in the sample are placed by 840 institutions during the period from January 1, 1999 to December 31, 2005. Panel A presents descriptive statistics for the ANcerno institutional trading data for each year of our sample period. Panel B reports descriptive statistics for stocks traded by ANcerno institutions. We obtain share price, total shares outstanding, stock returns, and trading volume from the CRSP stock database. Our sample includes only common stocks (those with a sharecode of 10 or 11 in CRSP). We obtain book value of equity from Compustat. Quoted spreads are from the TAQ database. Idiosyncratic volatility is the standard deviation of residuals for regressions of daily stock returns on market returns. Amihud illiquidity measure is constructed as the average of daily ratios between absolute return and dollar trading volume. We compute stock characteristics each quarter. Market capitalization and book-to-market ratio are as of the end of the previous quarter. All other stock characteristics are measured based on the 12-month period until the end of the previous quarter. We report the time-series mean, median, standard deviation, maximum, and minimum of quarterly cross-sectional averages for all stock characteristics.

Panel A: ANcerno Data							
	1999	2000	2001	2002	2003	2004	2005
Total number of funds	1,846	1,699	1,733	1,717	1,600	1,545	1,241
Total number of institutions	382	376	404	430	405	406	376
Median number of funds per inst.	4	4	4	4	4	3	3
Total number of stocks	6,150	5,906	5,082	4,692	4,736	4,927	4,763
Total number of trades (millions)	5.64	7.56	9.05	12.32	12.35	21.43	19.10
Total share volume (billion)	50.69	73.44	100.99	135.04	112.30	155.92	127.40
Total dollar volume (\$trillion)	2.25	3.20	3.06	3.23	2.76	4.46	3.95
Average share volume per trade	8,988	9,714	11,159	10,961	9,093	7,276	6,669
Median share volume per trade	1,700	1,500	1,400	1,300	1,050	700	453
Average dollar volume per trade	398,803	423,726	337,633	262,359	223,126	208,027	206,902
Median dollar volume per trade	60,030	54,970	39,200	30,300	27,297	20,568	14,232
Panel B: Stock Characteristics							
	Mean	Median	SD	Maximum	Minimum		
Market capitalization (\$billion)	2.67	2.76	0.34	3.28	2.03		
Book-to-market ratio	0.51	0.49	0.06	0.63	0.43		
Lagged 12-month return (%)	4.31	2.38	19.11	57.78	-24.83		
Turnover (%)	163.42	158.39	15.84	195.98	143.13		
Idiosyncratic volatility (%)	47.93	49.71	11.28	65.30	31.09		
Amihud illiquidity measure ($\times 10^4$)	0.22	0.15	0.14	0.58	0.09		

Table I: Descriptive Statistics for ANcerno Institutional Trading Data and Stock Characteristics

Table II
Performance of Intraquarter Round-Trip Trades

Institutional trading data are obtained from ANcerno Ltd., and the trades in the sample are placed by 3,816 funds during the period from January 1, 1999 to December 31, 2005. Of these funds, 227 are money manager funds and 3,589 are pension funds. The sample includes only common stocks (those with a sharecode of 10 or 11 in the CRSP database). We calculate the holding-period return for each round-trip trade as the percentage difference between the sell price and buy price. DGTW-adjusted return is the raw holding-period return less the DGTW benchmark return over the identical holding period. We calculate principal-weighted average returns across all intra-quarter round-trip trades for each fund and each quarter. We then take a simple average across all funds and quarters and report raw and DGTW abnormal returns both before and after commissions in Panel A. In Panel B, we divide all funds into five quintiles at the end of each quarter based on the DGTW adjusted returns for their round-trip trades. We then report the average DGTW adjusted returns for these quintiles during the quarter of portfolio formation and the subsequent four quarters. All returns are expressed in percent. Numbers in parentheses are t -statistics, which are computed based on two-way clustered standard errors.

Panel A: Performance of Round-Trip Trades					
	All	Pension Funds	Money Manager Funds		
Raw return	2.99 (4.79)	3.07 (4.70)	2.00 (4.23)		
Raw return (after commissions)	2.71 (4.30)	2.79 (4.24)	1.69 (3.64)		
DGTW adj. return	2.09 (3.81)	2.17 (3.84)	1.17 (2.28)		
DGTW adj. return (after commissions)	1.80 (3.27)	1.88 (3.32)	0.87 (1.71)		
% of intra-quarter round-trip trades	22.89	18.30	33.65		
Panel B: Performance Persistence of Round-Trip Trades					
Current Quarter DGTW adj. Return Quintiles	Quarters				
	Q+0	Q+1	Q+2	Q+3	Q+4
q1	-13.16 (-20.11)	-2.67 (-4.79)	-2.54 (-3.63)	-2.05 (-2.75)	-1.82 (-2.00)
q2	-2.98 (-8.77)	-0.20 (-0.38)	-0.23 (-0.46)	-0.32 (-0.50)	-0.21 (-0.34)
q3	0.94 (2.31)	1.28 (2.43)	1.40 (2.36)	1.29 (2.48)	1.32 (2.19)
q4	5.48 (9.00)	3.74 (5.02)	3.51 (4.33)	2.96 (3.89)	3.02 (4.66)
q5	20.15 (12.60)	6.85 (7.91)	6.89 (8.13)	6.10 (9.24)	6.09 (6.47)
q5-q1	33.31 (17.08)	9.53 (11.37)	9.43 (11.99)	8.15 (10.23)	7.91 (7.10)

Table II: Performance of Intraquarter Round-Trip Trades

7.2 Table III: Interim Trading Performance of Institutional Investors + Table IV: Persistence of Interim Trading Performance

Read:

- Table III reports the main all-trade result. Equal-weighted buy-minus-sell DGTW performance is 0.74%; after commissions it is 0.34%.
- Principal-weighted buy-minus-sell performance is 0.57%; after commissions it is 0.27%.
- Buy trades drive much of the result: buys outperform benchmarks, while sells only weakly underperform.
- Money managers have higher principal-weighted after-commission performance than pension funds, but both groups show positive skill.
- Table IV sorts funds by current-quarter interim performance and shows strong persistence for high-performance funds.
- Top-quintile funds have future buy-minus-sell performance of 1.56%, 1.16%, 1.33%, and 1.10% over the next four quarters.

Detailed interpretation: Table III is the central skill table because it uses all trades rather than only round trips. The after-commission numbers are smaller but remain significant, meaning the result is not an artifact of ignoring brokerage costs. Table IV strengthens the skill interpretation because past winners remain future winners. Persistence concentrated among high-skill funds is more consistent with skill than with temporary bad luck among losers.

Economic message: Institutions generate economically meaningful interim alpha. The important result is not just positive average performance, but persistent cross-sectional differences in short-horizon trading skill.

Table III
Interim Trading Performance of Institutional Investors

Institutional trading data are obtained from ANcerno Ltd. and the trades in the sample are placed by 3,816 funds during the period from January 1, 1999 to December 31, 2005. Of these funds, 227 are money manager funds and 3,589 are pension funds. Money manager funds are responsible for more than half of the dollar value of trading volume in the sample. The sample includes only common stocks (those with a sharecode of 10 or 11 in the CRSP database). For each trade, we calculate the raw cumulative stock return from the execution price until the end of the quarter. We adjust the raw cumulative return by the DGTW benchmark return over the same period. For each fund in each quarter, we then compute the equal-weighted or principal-weighted DGTW adjusted returns separately for buys and sells. We take the difference in DGTW adjusted returns between buys and sells. We report a simple average across all funds and quarters. Panel A presents equal-weighted DGTW adjusted returns, and Panel B presents principal-weighted DGTW adjusted returns. All returns are expressed in percent. Numbers in parentheses are *t*-statistics, which are computed based on two-way clustered standard errors.

	All	Pension Funds	Money Manager Funds
Panel A: Equal-Weighted DGTW Adjusted Return			
Buy	0.67 (4.04)	0.68 (3.96)	0.50 (4.38)
Sell	-0.06 (-0.43)	-0.06 (-0.40)	-0.13 (-1.06)
Buy-Sell	0.74 (7.73)	0.74 (7.51)	0.63 (6.28)
Buy-Sell (after commissions)	0.34 (3.26)	0.34 (3.16)	0.27 (2.87)
Panel B: Principal-Weighted DGTW Adjusted Return			
Buy	0.54 (3.57)	0.55 (3.48)	0.45 (4.22)
Sell	-0.03 (-0.18)	-0.01 (-0.08)	-0.26 (-1.73)
Buy-Sell	0.57 (5.73)	0.56 (5.52)	0.72 (4.75)
Buy-Sell (after commissions)	0.27 (2.67)	0.26 (2.52)	0.40 (2.74)

Table III: Interim Trading Performance of Institutional Investors

Table IV
Persistence of Interim Trading Performance

Institutional trading data are obtained from ANcerno Ltd. and the trades in the sample are placed by 3,816 funds during the period from January 1, 1999 to December 31, 2005. The sample includes only common stocks (those with a sharecode of 10 or 11 in the CRSP database). For each trade, we calculate the raw cumulative stock return from the execution price until the end of the quarter. We adjust this cumulative return by the DGTW benchmark return over the same period. For each fund in each quarter, we then compute the principal-weighted DGTW adjusted returns separately for buys and sells. We take the difference in DGTW adjusted returns between buys and sells. At the end of the each quarter, we divide all funds into five quintiles based on the principal-weighted DGTW adjusted returns for buys minus sells. We then report the average DGTW adjusted returns for these quintiles during the quarter of portfolio formation and for the subsequent four quarters. All returns are expressed in percent. Numbers in parentheses are *t*-statistics, which are computed based on two-way clustered standard errors.

Current Quarter Performance Quintiles		Quarters				
		Q+0	Q+1	Q+2	Q+3	Q+4
q1	Buy	-3.88 (-14.12)	0.15 (0.95)	0.13 (0.63)	0.21 (1.23)	0.11 (0.57)
	Sell	4.79 (10.70)	0.48 (2.38)	0.38 (1.81)	0.44 (2.48)	0.27 (1.26)
	Buy-Sell	-8.67 (-15.11)	-0.33 (-1.49)	-0.24 (-1.38)	-0.23 (-1.14)	-0.16 (-0.66)
q2	Buy	-0.78 (-8.16)	0.36 (3.73)	0.29 (2.73)	0.21 (1.36)	0.30 (2.35)
	Sell	1.05 (6.98)	0.05 (0.33)	-0.08 (-0.61)	-0.01 (-0.07)	0.11 (0.64)
	Buy-Sell	-1.84 (-11.59)	0.31 (2.50)	0.37 (3.01)	0.22 (1.75)	0.18 (1.12)
q3	Buy	0.37 (3.60)	0.36 (2.63)	0.50 (4.04)	0.46 (2.54)	0.36 (3.93)
	Sell	-0.25 (-2.45)	-0.14 (-1.11)	-0.15 (-1.08)	0.02 (0.10)	-0.16 (-1.06)
	Buy-Sell	0.62 (10.06)	0.50 (5.04)	0.65 (4.97)	0.44 (3.76)	0.52 (3.69)
q4	Buy	1.72 (12.58)	0.59 (3.44)	0.54 (3.38)	0.64 (3.82)	0.59 (3.73)
	Sell	-1.41 (-14.18)	-0.15 (-0.70)	-0.17 (-0.86)	-0.30 (-1.40)	-0.36 (-2.64)
	Buy-Sell	3.13 (20.29)	0.74 (5.18)	0.70 (4.82)	0.93 (6.42)	0.96 (7.41)
q5	Buy	5.29 (10.35)	1.13 (3.55)	0.96 (3.45)	1.13 (3.44)	0.63 (2.56)
	Sell	-4.31 (-15.80)	-0.44 (-2.18)	-0.20 (-0.95)	-0.20 (-0.88)	-0.46 (-2.61)
	Buy-Sell	9.60 (16.19)	1.56 (5.69)	1.16 (5.73)	1.33 (6.40)	1.10 (6.12)
q5-q1	Buy	9.17 (15.21)	0.97 (3.74)	0.82 (4.18)	0.91 (2.87)	0.52 (2.49)
	Sell	-9.11 (-16.02)	-0.92 (-4.86)	-0.58 (-3.19)	-0.64 (-3.53)	-0.74 (-3.39)
	Buy-Sell	18.27 (16.26)	1.89 (5.16)	1.41 (5.55)	1.56 (5.47)	1.26 (5.19)

Table IV: Persistence of Interim Trading Performance

7.3 Table V: Implied Quarterly Trading Performance of Institutional Investors, Parts 1 and 2

Read:

- Table V reconstructs what the results would look like if the authors ignored high-frequency trading data and used implied quarterly trades.
- Panel A shows that equal-weighted buy-minus-sell implied quarterly performance is -0.33%; principal-weighted performance is -0.24%.
- The quarterly proxy therefore suggests no positive skill, and possibly negative skill.
- Panel B shows little evidence of persistence. The top-quintile implied quarterly portfolio does not continue to outperform the bottom quintile.
- The results sharply contrast with Tables III and IV, despite using the same underlying sample.

Detailed interpretation: Table V is the measurement-error table. It shows that the same institutions can look skilled with trade-level data and unskilled with quarterly net positions. The difference comes from

missing the exact timing of trades and netting out round-trip trading. This is a methodological warning for any study that uses quarterly holdings to measure short-horizon trading activity.

Economic message: The table demonstrates that low-frequency data can erase the evidence of skill. Measurement frequency is not a technical detail; it changes the economic conclusion.

Table V
Implied Quarterly Trading Performance of Institutional Investors

Institutional trading data are obtained from ANcerno Ltd., and the trades in the sample are placed by 3,816 funds during the period from January 1, 1999 to December 31, 2005. Of these funds, 227 are money manager funds and 3,589 are pension funds. The sample includes only common stocks (those with a sharecode of 10 or 11 in the CRSP database). For each fund, we track all trades within the quarter and calculate the net trading position for each stock as of the quarter-end. We then calculate the DGTW equal- and principal-weighted abnormal performance for buys (positive net trading imbalance) and sells (negative trading imbalance) over the subsequent quarter. Finally, we take the difference in DGTW adjusted returns between buys and sells. We report a simple average across all funds and quarters in Panel A. In Panel B, we divide all funds into five quintiles at the end of each quarter based on their principal-weighted implied quarterly trading performance. We then report the average implied quarterly trading performance for these quintiles during the quarter of portfolio formation and the subsequent four quarters. All returns are expressed in percent. Numbers in parentheses are *t*-statistics, which are computed based on two-way clustered standard errors.

Panel A: Implied Quarterly Trading Performance			
	All	Pension Funds	Money Manager Funds
Equal-weighted DGTW adj. Return			
Buy	-0.05 (-0.14)	-0.06 (-0.17)	0.20 (0.70)
Sell	0.28 (0.79)	0.28 (0.79)	0.27 (0.73)
Buy-Sell	-0.33 (-2.14)	-0.34 (-2.20)	-0.08 (-0.38)
Principal-weighted DGTW adj. Return			
Buy	-0.05 (-0.17)	-0.05 (-0.15)	-0.10 (-0.84)
Sell	0.19 (0.78)	0.20 (0.81)	0.10 (0.74)
Buy-Sell	-0.24 (-1.70)	-0.24 (-1.66)	-0.20 (-1.21)

Panel B: Persistence of Implied Quarterly Trading Performance					
Performance	Quarters				
Quintiles	Q+0	Q+1	Q+2	Q+3	Q+4
q1 (low)					
Buy	-7.23 (-10.74)	-0.03 (-0.07)	0.04 (0.07)	0.01 (0.02)	-0.03 (-0.08)
Sell	7.02 (11.76)	0.45 (1.17)	0.21 (0.53)	-0.01 (-0.04)	0.42 (0.99)
Buy-Sell	-14.24 (-15.37)	-0.48 (-1.71)	-0.18 (-0.61)	0.03 (0.07)	-0.46 (-1.19)

(continued)

Table V: Implied Quarterly Trading Performance, Part 1

Table V—Continued

Panel B: Persistence of Implied Quarterly Trading Performance					
Performance	Quarters				
Quintiles	Q+0	Q+1	Q+2	Q+3	Q+4
q2					
Buy	-2.25 (-7.75)	-0.17 (-0.50)	-0.40 (-1.00)	0.04 (0.10)	0.16 (0.42)
Sell	1.94 (7.56)	-0.04 (-0.11)	0.02 (0.09)	0.35 (1.02)	0.15 (0.48)
Buy-Sell	-4.19 (-15.72)	-0.13 (-0.66)	-0.43 (-1.63)	-0.03 (-1.33)	0.01 (0.03)
q3					
Buy	-0.08 (-0.32)	-0.07 (-0.23)	0.11 (0.47)	0.11 (0.35)	-0.09 (-0.28)
Sell	0.12 (0.52)	0.19 (0.84)	0.31 (1.22)	0.19 (0.82)	0.25 (0.79)
Buy-Sell	-0.20 (-1.90)	-0.25 (-1.48)	-0.20 (-0.99)	-0.08 (-0.35)	-0.34 (-1.72)
q4					
Buy	1.97 (6.45)	-0.03 (-0.13)	0.05 (0.20)	0.07 (0.22)	0.11 (0.33)
Sell	-1.90 (-7.05)	0.11 (0.54)	0.33 (1.60)	0.23 (0.81)	0.29 (1.18)
Buy-Sell	3.87 (12.94)	-0.14 (-0.81)	-0.28 (-1.18)	-0.16 (-0.69)	-0.17 (-0.74)
q5 (high)					
Buy	7.33 (10.79)	0.04 (0.11)	0.37 (0.88)	0.04 (0.13)	0.16 (0.35)
Sell	-6.21 (-13.06)	0.03 (0.08)	0.19 (0.47)	0.43 (1.38)	-0.13 (-0.37)
Buy-Sell	13.54 (15.77)	0.01 (0.04)	0.18 (0.47)	-0.38 (-1.49)	0.29 (0.95)
q5-q1					
Buy	14.56 (14.24)	0.07 (0.17)	0.33 (0.57)	0.03 (0.08)	0.20 (0.51)
Sell	-13.23 (-15.96)	-0.42 (-1.27)	-0.02 (-0.08)	0.44 (1.13)	-0.55 (-1.37)
Buy-Sell	27.78 (16.07)	0.49 (1.05)	0.36 (0.68)	-0.41 (-1.00)	0.75 (1.72)

Table V: Implied Quarterly Trading Performance, Part 2

7.4 Table VI: Stock Characteristics and Interim Trading Performance, Parts 1 and 2

Read:

- Table VI asks where interim trading skill is strongest.
- Performance is positive in both small and large stocks, but the difference is not the main result.
- High idiosyncratic volatility stocks have much stronger buy-minus-sell performance than low-IVOL stocks.
- High illiquidity stocks and high quoted-spread stocks also show larger interim performance.
- Book-to-market, lagged return, and turnover sorts are less central; the strongest mechanism evidence is in IVOL and liquidity-related variables.

Detailed interpretation: The mechanism is consistent with informed trading in environments where public information is weaker or arbitrage is more limited. High IVOL and illiquidity make mispricing harder to arbitrage and make private information more valuable. However, these stocks are also expensive to trade, so the result implies the information advantage is large enough to cover costs.

Economic message: Skill is concentrated where information advantages should be most valuable. This helps distinguish genuine information-based skill from generic risk exposure.

Table VI
Stock Characteristics and Interim Trading Performance

Institutional trading data are obtained from ANcerno Ltd., and the trades in the sample are placed by 3,816 funds during the period from January 1, 1999 to December 31, 2005. Of these funds, 227 are money manager funds and 3,589 are pension funds. The sample includes only common stocks (those with a sharecode of 10 or 11 in the CRSP database). We assign stocks to size, book-to-market, lagged return, turnover, idiosyncratic volatility, illiquidity, and quoted spread deciles based on NYSE breakpoints. All of these variables are based on data available at the end of the previous quarter. Lagged return, turnover, idiosyncratic volatility, illiquidity, and quoted spread are calculated using 12 months of data ending at the previous quarter's end. The decile portfolio with the smallest value of the sorting variable is assigned to decile 1; the decile portfolio with the largest value of the sorting variable is assigned to decile 10. In Panel A, we report the average decile values for buys and sells separately and also for pension funds and money manager funds separately. For each stock characteristic, we group all stocks into two categories, those below the NYSE median and those above the NYSE median. We calculate the abnormal trading performance for each stock characteristic category as follows. We calculate the raw cumulative stock return for each trade from the execution price until the end of the quarter. We adjust this raw cumulative return by the DGTW benchmark return over the same period. For each fund in each quarter, we then compute the principal-weighted DGTW adjusted returns separately for buys and sells. Finally, we take the difference in DGTW adjusted returns between buys and sells. In Panel B, we report the simple average abnormal trading performance measure for each stock characteristic category across all funds and quarters. All returns are expressed in percent. Numbers in parentheses are *t*-statistics, which are computed based on two-way clustered standard errors.

Panel A: Stock Characteristics				
	All	Pension Funds	Money Manager Funds	
Size Deciles	Buy	7.98	7.99	7.82
	Sell	8.09	8.10	7.91
Book-to-Market Deciles	Buy	3.60	3.59	3.83
	Sell	3.67	3.65	3.92
Lagged Return Deciles	Buy	5.87	5.89	5.56
	Sell	6.07	6.09	5.72
Turnover Deciles	Buy	7.04	7.05	6.87
	Sell	6.97	6.98	6.83
Idiosyncratic Volatility Deciles	Buy	5.60	5.61	5.39
	Sell	5.43	5.44	5.31
Illiquidity Deciles	Buy	2.80	2.79	2.92
	Sell	2.72	2.71	2.87
Quoted Spread Deciles	Buy	3.46	3.45	3.65
	Sell	3.54	3.53	3.73

Panel B: Abnormal Trading Performance by Stock Characteristics				
	Average Decile	Buy	Sell	Buy-Sell
Small stocks	1.88	1.17 (3.45)	0.35 (1.23)	0.82 (5.07)
Large stocks	7.83	0.35 (2.02)	-0.24 (-1.55)	0.59 (5.45)
Small-Large				0.23 (1.22)

Table VI: Stock Characteristics and Interim Trading Performance, Part 1

Table VI—Continued

Panel B: Abnormal Trading Performance by Stock Characteristics				
	Average Decile	Buy	Sell	Buy-Sell
Growth stocks	2.75	0.43 (2.51)	-0.15 (-0.97)	0.58 (5.88)
Value Stocks	8.29	1.34 (3.56)	0.50 (1.50)	0.84 (5.02)
Growth-Value				-0.26 (-1.42)
Past losers	2.55	0.35 (1.68)	-0.15 (-0.64)	0.50 (3.19)
Past winners	8.34	0.61 (3.55)	-0.11 (-0.57)	0.71 (6.94)
Losers-Winners				-0.21 (-1.09)
Low turnover stocks	2.38	0.80 (2.56)	0.06 (0.19)	0.74 (6.93)
High turnover stocks	8.54	0.55 (2.08)	-0.08 (-0.35)	0.63 (5.01)
Low-High				0.11 (0.66)
Low IVOL stocks	3.08	0.56 (2.43)	0.11 (0.47)	0.45 (5.06)
High IVOL stocks	9.03	0.79 (2.16)	-0.09 (-0.24)	0.88 (5.58)
Low-High				-0.43 (-2.52)
Low illiquidity stocks	3.16	0.35 (2.31)	-0.15 (-1.05)	0.49 (4.65)
High illiquidity stocks	9.07	1.82 (4.92)	0.53 (1.83)	1.29 (4.26)
Low-High				-0.80 (-2.41)
Low quoted spread stocks	2.62	0.36 (2.43)	-0.21 (-1.27)	0.56 (5.22)
High quoted spread stocks	8.61	1.59 (5.39)	0.55 (2.14)	1.05 (6.10)
Low-High				-0.48 (-2.79)

Table VI: Stock Characteristics and Interim Trading Performance, Part 2

7.5 Table VII: Interim Trading Performance and Implicit Trading Costs

Read:

- Table VII compares implicit trading costs across interim performance quintiles.
- The low-performance quintile has implicit cost of 0.051%.
- The high-performance quintile has implicit cost of 0.090%.
- The high-minus-low difference is 0.039% and statistically significant.
- High-skill funds therefore appear to pay more to execute trades, not less.

Detailed interpretation: If high interim performance were just payment for providing liquidity, high-skill funds should execute at favorable prices and incur lower trading costs. Instead, the table shows that high-performance funds pay higher costs, consistent with liquidity demand. This is exactly what one would expect if funds are acting quickly on private information.

Economic message: The abnormal performance is not simply compensation for being a liquidity provider. Skilled institutions seem to pay for immediacy because their information is valuable and short-lived.

Table VII
Interim Trading Performance and Implicit Trading Costs

Institutional trading data are obtained from ANcerno Ltd., and the trades in the sample are placed by 3,816 funds during the period from January 1, 1999 to December 31, 2005. The sample includes only common stocks (those with a sharecode of 10 or 11 in the CRSP database). We divide all funds into five quintiles at the end of each quarter based on their principal-weighted interim trading performance measure. We then calculate the average implicit trading costs for each fund as follows. For each buy trade, we calculate the implicit trading cost as execution price less the volume-weighted average price (VWAP) of the same trading day. For each sell trade, we calculate the implicit trading cost as the VWAP of the same trading day less the execution price. The implicit trading costs are then scaled by the execution price and expressed in percentage terms. For each fund in each quarter, we then compute the principal-weighted average implicit trading cost. We then report the average implicit trading costs for each abnormal trading performance quintile. Numbers in parentheses are *t*-statistics, which are computed based on two-way clustered standard errors.

Interim Trading Performance Quintiles	Average Interim Trading Performance (%)	Implicit Trading Costs (%)
Q1 (low)	-8.67	0.051 (5.87)
Q2	-1.84	0.058 (8.21)
Q3	0.62	0.056 (9.32)
Q4	3.13	0.070 (8.71)
Q5 (high)	9.60	0.090 (7.42)
Q5-Q1	18.27	0.039 (3.82)

Table VII: Interim Trading Performance and Implicit Trading Costs

8 Result map

- **Result 1: Round-trip skill.** Institutions earn high DGTW-adjusted abnormal returns on intraquarter round-trip trades.
- **Result 2: All-trade interim skill.** Buy trades outperform sell trades from execution to quarter-end after benchmark adjustment and commissions.
- **Result 3: Skill persistence.** Current high-skill funds continue to outperform in later quarters.
- **Result 4: Quarterly-data failure.** Net quarterly trades suggest negative or insignificant performance, showing that low-frequency data miss interim skill.
- **Result 5: Information environments.** Performance is strongest in high-IVOL and illiquid stocks.
- **Result 6: Liquidity-demand evidence.** Skilled funds have higher implicit costs, inconsistent with a simple liquidity-provision explanation.
- **Result 7: Economic magnitude.** Average skill contributes 20–26 basis points annually; top-skill funds may generate over 1% per year from interim trading.

9 Short summary

- The paper uses proprietary ANcerno transaction data to study institutional trading skill at the trade level.
- Standard quarterly holdings data miss round-trip trades and mismeasure trade timing.
- Round-trip trades are profitable and performance is persistent.
- All-trade interim performance is positive after DGTW adjustment and commissions.
- Persistence is driven by high-skill funds rather than by chronic underperformers.
- Implied quarterly trading performance is negative or insignificant, showing that quarterly data understate short-horizon skill.
- Skill is strongest in stocks with high idiosyncratic volatility, low liquidity, and high quoted spreads.

- High-skill funds pay higher implicit trading costs, suggesting they demand liquidity to exploit information.
- The results support the idea that informed institutions earn abnormal returns when information acquisition and rapid trading are costly.

10 Conclusion

10.1 Core conclusion

Puckett and Yan show that institutional investors possess measurable interim trading skills when their actual trades are observed. Institutional buys outperform sells within the trading quarter, and this performance persists across funds. The evidence is strongest when the authors use the exact execution date and price of each trade.

10.2 Economic intuition

If institutional investors trade on short-lived private information, the performance window is short. A fund may buy a stock, profit from rapid price adjustment, and sell before quarter-end. Quarterly holdings data would record little or no change in holdings and therefore miss the trade entirely. The economic value of the ANcerno data is that they observe the decision at the frequency at which the information advantage operates.

10.3 Methodological implication

The paper’s key methodological lesson is that holdings-based proxies can be misleading for short-horizon questions. The same underlying institutions look skilled using trade-level interim measures and unskilled using quarterly implied trades. Future work on institutional trading around earnings announcements, analyst recommendations, mergers, and other events should therefore use high-frequency trade data when possible.

10.4 Asset-pricing implication

The results are consistent with a Grossman–Stiglitz view of markets: information acquisition is costly, and informed traders must earn abnormal returns to justify those costs. The fact that performance is strongest in high-IVOL and illiquid stocks suggests that information advantages survive where arbitrage is difficult.

10.5 Policy implication

The findings speak to disclosure policy. More frequent institutional portfolio disclosure may improve transparency, but it may also erode the short-term information advantages through which institutions earn returns. If institutions are compensated for costly information production, forcing more frequent disclosure could reduce their incentives to acquire and trade on information.

10.6 Limitations

The paper does not observe management fees or actual investor-level fund returns for the full sample. Therefore, it cannot determine whether interim trading gains accrue to fund investors or are captured by managers through fees. The sample is also based on ANcerno clients, which are larger than average institutions. The authors provide evidence that this does not drive the main results, but external validity remains a natural caveat.

10.7 Final takeaway

Institutional trading skill exists, is persistent, and is concentrated where information advantages are most valuable. But one only sees it at the right frequency. Quarterly holdings data can hide exactly the trades that reveal skill.